EXTRA PRACTICE QUESTIONS + TOP-30 LIST

> ### 🎯 Claude Predicted Practice Questions
>
> **These are predicted/practice questions based on syllabus and emphasis, not leaked or guaranteed questions.** They are *not* copied from your assignments. Each has a pointer (→) to the section that contains the answer. Numerical answers are given at the end.

PART A — 30 SHORT QUESTIONS (1–2 marks)

Claude Predicted Practice Question — each one.

1. Why is "no shared memory" a defining property of a distributed system? → 01 §1
2. Give two examples of heterogeneity in a distributed system and how middleware handles it. → 01 §2
3. What is location transparency? Give an example. → 01 §2
4. Why is a server called a single point of failure? → 01 §3.1
5. What is a DHT in P2P? (one line) → 01 §3.2
6. Name two real-world grid computing projects. → 01 §3.3
7. List three characteristics of cloud computing. → 01 §3.4
8. What is horizontal scalability? → 01 §4
9. What is meant by "openness" of a distributed system? → 01 §2
10. What is jitter and where does it matter? → 01 §5.3
11. Why can't timeouts detect failures reliably in an asynchronous system? → 01 §5.3
12. Give one threat to a channel in the security model. → 01 §5.3
13. Which device joins two different networks? → 01 §6
14. Why is UDP used for video streaming? → 01 §6
15. What is a socket? → 01 §7.1
16. What is a race condition in shared memory? → 01 §7.2
17. Why can pointers not be passed in RPC? → 01 §10
18. What is an IDL? → 01 §10
19. What is the difference between `bind` and `rebind` in Java RMI? → 01 §12.7
20. What does `UnicastRemoteObject` do? → 01 §12.3
21. What is a topic-based subscription? → 01 §11
22. What is a logical clock? → 02 §5
23. What is a "concurrent event"? Give an example. → 02 §4
24. Why must the receive rule of Lamport clock use `max`? → 02 §5
25. What is UTC? → 02 §7
26. What is a stable predicate? Give an example. → 02 §10
27. What are the three types of events in a process? → 02 §3
28. Why is starvation a problem in mutual exclusion? → 02 §12
29. What is the difference between delivering and receiving a multicast message? → 02 §14

30. State the FLP impossibility result in one line. → 02 §15

PART B — 20 MEDIUM QUESTIONS (3–4 marks)

Claude Predicted Practice Question — each one.

1. Explain the three major challenges caused by the absence of a global clock in a distributed system. → 02 §1, §7
2. "A distributed system is more than a set of networked computers." Justify with reference to transparency and coordination. → 01 §1–2
3. How is cloud computing different from grid computing? Explain with examples. → 01 §3.5
4. Explain the role of middleware in masking heterogeneity. → 01 §5.2
5. Explain caching and proxy servers as architectural elements. → 01 §5.2
6. Explain mobile code and mobile agents with advantages and security concerns. → 01 §5.2
7. Explain the failure model and the notion of masking failures. → 01 §5.3
8. Describe how a synchronous system differs from an asynchronous one in designing failure detectors. → 01 §5.3
9. Why is the request–reply protocol central to RPC and RMI? Explain marshalling and semantics. → 01 §7, §9
10. Explain the three invocation semantics and state which is used by Java RMI and why idempotent operations matter. → 01 §9
11. Why are remote exceptions necessary in RMI but not for local calls? → 01 §8
12. Explain event-based (publish–subscribe) architecture and its advantages over polling. → 01 §11
13. Distinguish between logical clocks and vector clocks using an example where Lamport clocks fail to detect concurrency. → 02 §5–6
14. Explain how skew is bounded by drift rate and how often clocks must be synchronised. → 02 §7
15. Explain Berkeley algorithm with an example of three machines. → 02 §7
16. Why is a consistent cut needed for recording a global state? Give an example of an inconsistent cut. → 02 §9
17. What is "Possibly Φ" and "Definitely Φ" in distributed debugging? → 02 §10
18. Explain why Ricart–Agrawala guarantees mutual exclusion and freedom from starvation. → 02 §12
19. What happens in the Ring election algorithm if two processes start elections at the same time? → 02 §13
20. Explain causal ordering vs total ordering in multicast with an example each. → 02 §14

PART C — 15 LONG-ANSWER QUESTIONS (7–10 marks)

Claude Predicted Practice Question — each one.

1. Define distributed system. Discuss its characteristics, advantages and disadvantages, and the main design challenges. → 01 §1–4
2. Discuss in detail the architectural models of distributed systems and explain how middleware supports them. → 01 §5.2
3. Explain the interaction, failure and security models. Which assumptions does each make about the network? → 01 §5.3
4. Explain the TCP/IP protocol stack and how internetworking supports distributed applications. → 01 §6
5. Explain interprocess communication using sockets (UDP and TCP), marshalling and the request–reply protocol. → 01 §7
6. Explain RPC in detail: components, steps, failure semantics and issues. Compare with RMI. → 01 §10
7. Explain the complete architecture and working of Java RMI with a student-database example. → 01 §12
8. Explain events and notifications with the publish–subscribe model, roles, subscription types and applications. → 01 §11
9. Discuss physical clock synchronisation algorithms (Cristian, Berkeley, NTP) with formulas and examples. → 02 §7
10. Explain happened-before, Lamport clocks and vector clocks; work out an example with 3 processes and 8 events. → 02 §4–6
11. Explain global states, cuts, consistency and the Chandy–Lamport algorithm; prove that the recorded state is consistent. → 02 §9
12. Discuss distributed debugging: predicates, lattice of global states, possibly/definitely, and monitor process design. → 02 §10
13. Compare all mutual exclusion algorithms on messages, delay and failure tolerance. → 02 §12
14. Discuss multicast communication: reliable multicast properties, FIFO/causal/total order and implementation. → 02 §14
15. Explain consensus, Byzantine generals and interactive consistency; state bounds on number of faulty processes. → 02 §15

PART D — 10 APPLICATION / CASE-STUDY QUESTIONS

Claude Predicted Practice Question — each one.

1. **Online ticket booking:** many users book the last seat simultaneously. Which distributed-system characteristics and which mutual-exclusion mechanism apply? → 01 §2, 02 §12
2. **YouTube-like service:** identify architectural elements (proxy/caches, multiple servers, CDN) and fundamental-model issues (latency, failures). → 01 §5
3. **Chat application:** choose message passing vs shared memory, TCP vs UDP; justify. → 01 §6–7
4. **University ERP using Java RMI:** which interfaces and objects are remote? How does the client find the service? What invocation semantics? → 01 §12
5. **Stock-price alerts to 10,000 traders:** design using publish–subscribe; discuss reliability and ordering. → 01 §11, 02 §14

6. **Distributed logging across servers:** how would you order events from servers with drifting clocks? Compare NTP with logical clocks. → 02 §5–8
7. **Bank with branches:** design a method to compute total money at a consistent instant while transfers continue. → 02 §9
8. **Cluster with a master node** that crashes every few days: choose an election algorithm and justify; discuss the effect of a flapping high-ID node. → 02 §13
9. **Replicated database with 7 replicas, 2 may be faulty:** can consensus be reached (crash vs Byzantine)? How many rounds? → 02 §15
10. **Group chat with total ordering:** ensure every member sees messages in the same order; describe the sequencer method and its weakness. → 02 §14

PART E — 10 NUMERICAL / PROBLEM-BASED QUESTIONS

Claude Predicted Practice Question — each one. (Answers below.)

N1. P1: a, b(send→P2), c. P2: d, e(recv from P1), f(send→P3). P3: g, h(recv from P2). All clocks start at 0. Give Lamport timestamps. N2. P2's clock reads 8; it receives a message with timestamp 8. Then P2 sends a message. What timestamps result? N3. Vector clocks (3 processes): P1 sends message with V = (2,0,0). P2 currently has V = (0,3,1). What is P2's clock after receiving? Does the send happen-before an event of P2 with V = (0,3,1)? N4. Cristian: T0 = 09:15:00.000, T1 = 09:15:00.400, server time = 09:15:07.100, minimum one-way delay = 0.05 s. Find the corrected time and accuracy. N5. Berkeley: master 10:00, slave A 09:58, slave B 10:07, slave C 10:03 (master included). Compute adjustments. N6. Two clocks each drift at most 5×10^{-6} s/s. After 10 minutes since sync, what is the maximum skew? N7. Compare number of messages for 10 processes: central server, Ricart–Agrawala, Lamport ME, ring (best/worst). N8. Bully algorithm: 8 processes, coordinator P8 crashes, process P7 detects. How many messages? What if P1 detects (count roughly)? N9. Ring election with N = 10, initiator is the anticlockwise neighbour of the highest-id process. Total messages? Best case? N10. Chandy–Lamport: balances P1 = 400, P2 = 350, P3 = 250. Before snapshot: P1 sends 30 to P2. P1 starts snapshot after sending. P3 sends 10 to P1 before receiving marker. FIFO channels. Find recorded local states and channel states, verify total.

Answers to numericals

N1. a=1, b=2 (msg ts 2); P2: d=1, e=max(1,2)+1=3, f=4 (ts 4); P3: g=1, h=max(1,4)+1=5. c (P1) = 3. **N2.** Receive: max(8,8)+1 = **9**; send next: **10**. **N3.** Receive: P2 first takes max((0,3,1), (2,0,0)) = (2,3,1), then own increment → **(2,4,1)**. Send event with (2,0,0) < (0,3,1)? Compare: 2 > 0 in first component ⇒ not ≤ ⇒ **not comparable ⇒ concurrent**, so no happened-before. **N4.** RTT = 0.4 s → time = 7.100 + 0.2 = **09:15:07.300**. Accuracy = ±(0.2 − 0.05) = **±0.15 s**. **N5.** Offsets: master 0, A −2, B +7, C +3. Avg = (0 −2 +7 +3)/4 = **+2 min** → target 10:02. Adjustments: master +2; A +4; B −5; C −1. **N6.** Max skew = $2\rho\Delta t = 2 \times 5\times10^{-6} \times 600$ = **6 ms**. **N7.** N = 10: central = 3; RA = 2×9 = **18**; Lamport = 3×9 = **27**; ring = 1 to 9 messages per entry (plus continuous token circulation). **N8.** P7 detects: ELECTION to P8 (1); P8 silent; P7 wins after timeout; COORDINATOR to P1–P6 = 6 ⇒ **N−2 = 6 coordinator messages + 1 election = 7**. If P1 detects: ELECTION to P2–P8 (7), OK replies from P2–P7 (6), and each of P2–P7 starts its own election to higher ids… total is on the order of **N^2 ≈ 60+** (roughly 7+6+

(6+5+...+1 elections + replies) + 7 coordinator). **N9.** Worst case = 3N−1 = **29** messages (9 election to reach highest + 10 to circle back + 10 elected). Best case = 2N = **20**. **N10.** Sequence: P1 sends 30 to P2 (P1 = 370, m in C12). P1 records **S1 = 370**, sends markers on C12, C13, records C21, C31. P3 sends 10 to P1 (P3 = 240, message on C31 before its marker). P2 receives 30 (P2 = 380) **then** marker on C12 → records **S2 = 380**, C12 = {}. P3 gets marker → records **S3 = 240**, C13 = {}. P1 receives 10 from P3 while recording → **C31 = {10}**, then marker. Total = 370 + 380 + 240 + 10 = **1000** = 400 + 350 + 250 ✓.

🔥 TOP 30 QUESTIONS TO PREPARE FIRST

(Ranking is my judgement from the official syllabus + the emphasis in your two BPIT assignments. It is not a guarantee.) Priority: 🔥🔥🔥 very high, 🔥🔥 high, 🔥 medium. Marks = expected marks at typical BPIT/GGSIPU-style weightage.

#	Question	Unit	Topic	Exp. marks	Priority	Why important
1	Design a Java RMI application (calculator/student service): architecture, interface, server, client, registry, code, output	I	Java RMI	5–10	🔥🔥🔥	Assignment case study + official "Case study – Java RMI"
2	Explain distributed objects and RMI (remote object, interface, stub/proxy, steps)	I	Distributed objects / RMI	4–8	🔥🔥🔥	A1-3b; syllabus
3	RMI vs RPC	I	RMI/RPC	3–5	🔥🔥🔥	Standard comparison; both in syllabus
4	Message passing vs shared memory	I	IPC	3–5	🔥🔥🔥	A1-3a explicit comparison
5	Compare client–server, P2P, grid, cloud	I	Examples of DS	4–5	🔥🔥🔥	A1-1b + syllabus lists the four
6	Explain system models: architectural & fundamental (diagrams)	I	System models	5–10	🔥🔥🔥	A1-2a
7	Define DS and explain characteristics with examples	I	Characteristics	2–5	🔥🔥🔥	A1-1a; base for compulsory question
8	Networking & internetworking: role of protocols	I	Networking	3–5	🔥🔥	A1-2b
9	Explain RPC (stubs, steps, semantics)	I	RPC	3–5	🔥🔥	Syllabus topic; needed for RMI vs RPC
10	Events and notifications (publish–subscribe)	I	Events	3–5	🔥	Syllabus topic, not in assignment → short/medium likely

#	Question	Unit	Topic	Exp. marks	Priority	Why important
11	Advantages of distributed systems	I	Advantages	1–3	🔥	Syllabus; easy marks
12	Lamport logical clock: rules + timestamp numerical	II	Logical clocks	3–5	🔥🔥🔥	A2-1b; numerical topic
13	Happened-before relation with example	II	Events/order	2–3	🔥🔥🔥	A2-1b; foundation
14	Vector clocks vs Lamport	II	Logical time	3–4	🔥	Natural follow-up; tough-paper question
15	Need for physical clock sync; skew, drift, delay	II	Physical clocks	3–4	🔥🔥	A2-1c
16	Cristian & Berkeley algorithms (with numericals)	II	Physical clocks	3–5	🔥🔥	Standard numericals; supports A2-1c
17	Logical vs physical time table	II	Time	2–3	🔥🔥	Standard compare question
18	Events and process states; role in synchronisation	II	Sync intro	2	🔥🔥	A2-1a
19	Global state: components and importance of consistency	II	Global state	3–4	🔥🔥🔥	A2-2a
20	Distributed snapshot (Chandy–Lamport) with P1, P2, P3	II	Global state	4–8	🔥🔥🔥	A2-2b case study
21	Consistent vs inconsistent cut (diagram)	II	Global state	2–3	🔥🔥	Ties to 19–20
22	Distributed debugging	II	Debugging	2–4	🔥	Syllabus; not in assignment
23	Requirements of mutual exclusion + any two algorithms	II	Mutual exclusion	4–8	🔥🔥🔥	A2-3a
24	Ricart–Agrawala algorithm with example & message complexity	II	Mutual exclusion	4–6	🔥🔥🔥	Most standard permission-based algorithm
25	Election: Bully algorithm with example	II	Election	4–6	🔥🔥🔥	A2-3b
26	Ring election algorithm with example	II	Election	4–6	🔥🔥🔥	A2-3b
27	Bully vs Ring comparison (messages, assumptions)	II	Election	3–5	🔥🔥🔥	A2-3b asks it explicitly
28	Consensus: definition, agreement, validity, termination, example	II	Consensus	3–5	🔥🔥🔥	A2-4a
29	Consensus with 5 processes, 1 may fail; failures & delays	II	Consensus	4–8	🔥🔥🔥	A2-4b case study

#	Question	Unit	Topic	Exp. marks	Priority	Why important
30	Multicast communication: reliable, FIFO/causal/total ordering	II	Multicast	3–5	🔥	Syllabus; likely short/medium

Study order suggestion: 1–7 (Unit-I core) → 12, 13, 19, 20 (Unit-II core numericals/diagram) → 23–29 (algorithms) → 15–18 → the 🔥 items.